Digital Security Alarm Using Logic Gates

Mini Digital Electronics Project • Circuit Simulation

1. Project Overview

A beginner-friendly digital logic project that models a simple security alarm. The alarm activates only when the security system is enabled and either the door or window sensor is active.

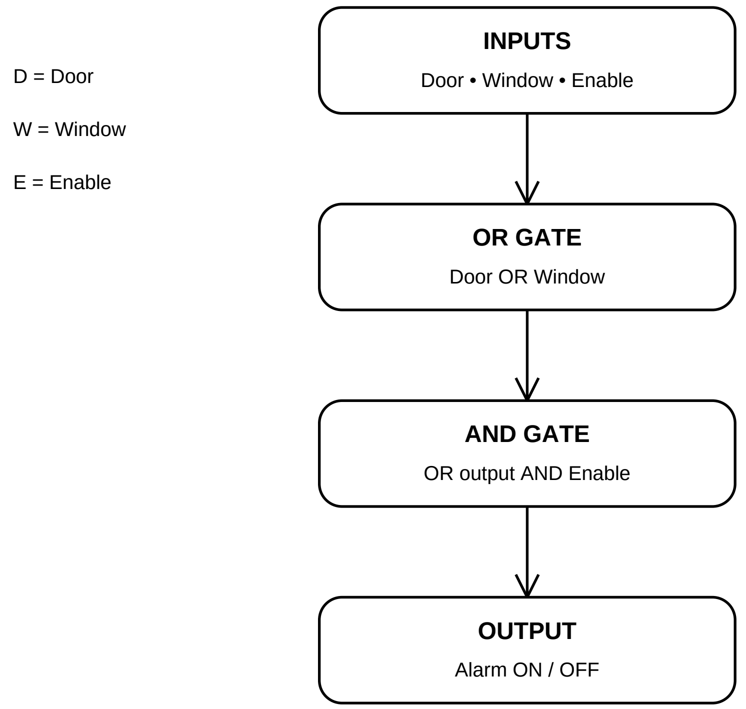
2. Objective

To understand logic gates, Boolean expressions, truth tables, inputs, outputs and digital circuit simulation by applying them to a simple real-world problem.

3. Inputs and Output

Symbol	Meaning	0 / 1
D	Door	Closed / Open
W	Window	Closed / Open
E	Enable	Security OFF / ON
A	Alarm	OFF / ON

4. Logic Flow Chart



Alarm = (Door OR Window) AND Enable

5. Logic Operation

Door and Window first enter an OR gate. If either is active, the OR output becomes 1. That output then enters an AND gate together with Enable. The alarm becomes 1 only when Enable is 1 and at least one sensor is 1.

6. Boolean Expression

$A = (D + W) \cdot E$

In Boolean algebra, + represents OR and · represents AND.

7. Truth Table

D	W	E	A
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

8. Demonstration

Add your own CircuitVerse screenshots here: complete circuit, Alarm OFF case, and Alarm ON case. These screenshots demonstrate that the circuit behaves according to the truth table.

9. Result

The Digital Security Alarm was successfully designed and simulated using OR and AND logic gates. The project shows how a real-world requirement can be converted into Boolean logic and then implemented as a digital circuit.

10. Learning Outcome

Learned the relationship between logic gates, Boolean expressions, truth tables and circuit behavior, while gaining practical experience with digital circuit simulation.

11. Future Scope

The design can later be upgraded with physical sensors, switches, LEDs, a buzzer and a microcontroller to create a real hardware prototype.